
When Personalisation Fails: Federated Portfolio Allocation Across Structurally Heterogeneous Markets

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Github Repository:

<https://github.com/umar-mk/dl-project>

Abstract

We study federated portfolio allocation across four structurally different international equity markets: S&P 500, FTSE 100, Nikkei 225, and KSE-100, each as a separate federated client holding 30 stocks. FedAvg improves over local-only training for all four markets, and the per-client gain correlates with a pre-training idiosyncrasy score ($r = +0.71$): more structurally independent markets benefit more from federation. SCAFFOLD reduces seed variance $8\times$ on S&P 500 and is bootstrap-significant over FedAvg on two markets ($p = 0.003$ and $p = 0.010$). Per-FedAvg’s meta-learned global model outperforms plain FedAvg at zero adaptation steps. However, eight personalisation methods spanning regularisation, meta-learning, mixture models, output interpolation, decoupled training, and similarity-weighted aggregation all fail to significantly improve any client over the best federation baseline. The optimal personalisation parameter collapses to fully-global or fully-local for every client and every method, with no market benefiting from intermediate adaptation. We connect this to the cross-silo personalisation phase transition of Liu et al. (NeurIPS 2022) and provide what we believe is the first empirical observation of this transition on real cross-market financial data.

1. Introduction

Portfolio allocation across international markets is a natural setting for federated learning. A manager in Karachi sees KSE-100 behaviour but not S&P 500 co-movements. A London-based manager sees FTSE 100 but not Nikkei 225 dynamics. A shared model trained across all four would capture broader market patterns. But raw financial data is legally protected and commercially sensitive. FL keeps data local and shares only model weight updates, making cross-market collaboration feasible without data disclosure.

Standard FL was designed for settings where clients differ in quantity but not in kind, for example hospitals with different patient volumes but similar patient populations. Financial markets break this assumption directly. The KSE-100 is a small, high-volatility, rupee-denominated emerging market. The S&P 500 is large, liquid, and dollar-denominated. Averaging their weight updates via FedAvg combines gradients from structurally different loss landscapes. Whether this helps or hurts each client, and by how much, is the first question we answer. Whether personalisation can correct any imbalance is the second.

Existing cross-market FL papers report only aggregate metrics. FQL (Cui et al., 2025) achieves a pooled Sharpe of 1.12 versus FedAvg’s 0.87 but does not break down which markets gain or lose. Yu et al. (Yu et al., 2022) demonstrate federated RL for portfolio management, but all 50 assets come from one market and one sector, so the federation is not genuinely cross-market. Neither paper asks whether specific clients are disadvantaged, and neither tests personalisation. Nosedá et al. (Nosedá et al., 2025) show FedAvg outperforms local-only training on S&P 500 under temporal non-IID splits, but again report only aggregate numbers without disaggregating by client.

Our study differs in three ways. We use four genuinely different markets from four different economies, currencies, and macro regimes. We report all results per client rather than pooled. And we test eight personalisation methods covering every major paradigm in the personalised FL literature.

Our contributions are: (1) a per-client measurement of FedAvg’s benefit across four heterogeneous markets, showing the gain is asymmetric and predicted by a pre-training idiosyncrasy score; (2) a systematic evaluation of eight personalisation methods showing that none significantly helps any client; (3) an empirical observation of the cross-silo personalisation phase transition of Liu et al. (Liu et al., 2022) on real financial data; (4) evidence that better aggregation (SCAFFOLD) and better training objectives (Per-FedAvg) outperform per-client adaptation in this regime; and (5) a capacity-matched Transformer experiment confirming the LSTM’s recurrent inductive bias is better suited to this task.

2. Related Work

Federated averaging and client drift. McMahan et al. (McMahan et al., 2017) introduced FedAvg and flagged non-IID data as an open problem from the start. Their own experiments showed a measurable gap between IID and non-IID performance, which later work confirmed grows worse with more local gradient steps. Karimireddy et al. (Karimireddy et al., 2020) proved this is caused by client drift: multiple local steps send each client’s model in a direction biased toward its own data, and the server average is then biased away from the true global optimum. SCAFFOLD corrects this with per-client control variates that track and subtract the bias during local training. Li et al. (Li et al., 2020) took a different approach: FedProx adds a proximal penalty to each client’s local objective that penalises deviation from the current global model, limiting how far any client can drift. Both methods are designed to improve the global model under heterogeneous data, not to give clients different models.

Personalised FL. The personalised FL literature starts from a different premise: rather than finding one good global model, find a global representation from which each client can quickly reach its own local optimum. Fallah et al. (Fallah et al., 2020) (Per-FedAvg) formalised this as MAML: the global model is trained so that one gradient step on any client’s data produces a good local model. Marfoq et al. (Marfoq et al., 2021) (FedEM) go further: they maintain M global component models and let each client learn soft mixing weights, so personalisation happens in mixture space rather than parameter space. Deng et al. (Deng et al., 2020) (APFL) take a simpler approach, interpolating the output of a global and a local model per client with a learned scalar weight. Li et al. (Li et al., 2021) (Ditto) train a separate per-client model alongside the global model with a proximal term anchoring it to the global. Ye et al. (Ye et al., 2023) (pFedGraph) personalise the aggregation step itself: each client aggregates a weighted sum of peer updates where the weights reflect gradient cosine similarity. This is qualitatively different from the others because it changes who each client listens to rather than how far it moves in parameter space.

Liu et al. (Liu et al., 2022) provide the theoretical context for our results. They show that in the cross-silo, small- K , same-task regime, the optimal regularisation strength for MR-MTL collapses to one of two extreme values: infinity (fully-global, plain FedAvg) or zero (fully-local), depending on the heterogeneity level. No intermediate value is optimal. This is the phase transition we observe empirically.

Heterogeneity characterisation. Jimenez G. et al. (Jimenez G. et al., 2024) provide a comprehensive taxonomy of non-IID types in FL, distinguishing feature distribution shift, label distribution skew, quantity imbalance, and spa-

tiotemporal skew. All four are present in our setting, with spatiotemporal skew being the dominant form: the four markets operate in different time zones, respond to different macro drivers, and have different crisis histories. The survey specifically highlights Hellinger Distance as a principled measure of distributional divergence in FL settings, which we use as one of two pre-training heterogeneity metrics.

FL for finance. Xu et al. (Xu et al., 2023) (FedPAC, ICLR 2023) separate the global model into a shared feature backbone and a per-client classification head. This architectural separation motivates our Split-FL experiment, which applies the same idea to portfolio allocation. Kennedy et al. (Kennedy et al., 2025) survey FL applications in finance and classify collaborative portfolio optimisation as a low-exposure FL application where institutions can co-train on local datasets while keeping proprietary trading data private. This motivates the practical setup of our study.

3. Methodology

3.1. Problem Formulation

Client i holds log returns for 30 stocks over 2010–2026. At each time step the model receives a 100-day window $X_t \in \mathbb{R}^{100 \times 180}$ and outputs portfolio weights $w_t \in \Delta^{30}$ (softmax over 30 assets, long-only). The training loss is the negative Sharpe ratio with a small turnover penalty:

$$\mathcal{L}(w, r) = -\frac{\mathbb{E}[e_t]}{\text{Std}(e_t)} \cdot \sqrt{252} + \lambda_{\text{TC}} |w_t - w_{t-1}|_1 \quad (1)$$

where $e_t = \sum_j w_{tj} r_{tj} - r_f$ is daily excess return, the standard deviation denominator is clipped at 10^{-4} to prevent gradient explosion in early training, and $\lambda_{\text{TC}} = 0.01$. Transaction costs of 10 bps one-way are applied at evaluation on actual weight changes, not included in the training penalty.

3.2. Data

Four markets serve as federated clients: S&P 500 (USD), FTSE 100 (GBP), Nikkei 225 (JPY), KSE-100 (PKR). Each holds 30 large-cap, liquid stocks selected using three fixed criteria: continuous price history from 2010 with minimal gaps, index relevance and liquidity, and no duplicate economic exposures (for example only GOOGL, not both GOOG and GOOGL). S&P 500, FTSE 100, and Nikkei 225 prices come from Yahoo Finance. KSE-100 prices are manually sourced because Yahoo Finance coverage of Pakistani equities is unreliable; each file requires comma-stripping and DD/MM/YYYY date handling.

Splits are strictly chronological: train 2010–2019, validation 2020–2021, test 2022–April 2026. The validation period covers COVID crash and recovery; the test period covers

the 2022 inflation shock, the 2023 recovery, and a late-cycle phase from 2024 onward. Six features per stock per day: log return, 5-day and 20-day rolling mean return, 5-day and 20-day rolling standard deviation, and 60-day cumulative return. Features are z-score normalised using training-period statistics, clipped at ± 5 , and the same statistics are applied identically to validation and test. No future information enters any feature.

We initially ran all experiments with 15 stocks per client. Personalisation methods failed uniformly. We expanded to 30 stocks to rule out thin cross-sectional signal as the explanation before concluding the failure was structural. The failure persisted.

3.3. Heterogeneity Metrics

We compute two pre-training heterogeneity scores on the training period only. Hellinger Distance (HD) between pairwise market-level return distributions, following Jimenez G. et al. (Jimenez G. et al., 2024): $HD(p, q) = \sqrt{\frac{1}{2} \int (\sqrt{p} - \sqrt{q})^2}$, estimated via kernel density. The *idiosyncrasy score*: $1 - R^2$ from regressing each market’s equal-weight daily return on the equal-weight mean of all four markets. This captures how much of each market’s return is structurally independent of the federation average. As Table 1 shows, these two metrics give different orderings: HD is dominated by volatility differences (Nikkei 225 scores highest), while idiosyncrasy captures structural independence (KSE-100 scores highest). Our results show idiosyncrasy is the better predictor of FL benefit.

Table 1. Pre-training heterogeneity scores (training period only). Idiosyncrasy = $1 - R^2$ against the equal-weight federation mean return.

Market	HD Score	Idiosyncrasy
KSE-100	0.073	0.750
S&P 500	0.082	0.591
FTSE 100	0.092	0.505
Nikkei 225	0.113	0.536

3.4. Architecture and Training

All methods use a 2-layer LSTM (hidden=96, dropout=0.1) with a linear softmax head: 184K parameters total. Input shape $(B, 100, 180)$, output $(B, 30)$. Training uses Adam with $lr = 5 \times 10^{-4}$, gradient clipping at norm 1.0, batch size 128, 5 local epochs per round, patience 15 rounds on aggregate validation Sharpe, and up to 150 rounds maximum. All experiments use 10 random seeds (42–51).

Ablations. We ran a lookback window ablation (20, 50, 100 days) and a FedProx μ ablation (0.001, 0.01, 0.1), both using seed 42. Window=100 strictly dominates: mean

Sharpe 0.810 versus 0.746 for window=50 and 0.676 for window=20. All main experiments use window=100. For FedProx, all three μ values produce mean Sharpe within 0.0005 of each other at all four markets, confirming the Sharpe loss dominates the proximal penalty regardless of strength.

3.5. Personalisation Methods

Non-FL baselines: equal-weight portfolio (B1), rolling Markowitz mean-variance (B2, 60-day estimation window), local-only LSTM (B3), centralised LSTM trained on pooled data from all markets (B4).

Standard FL baselines: FedAvg (McMahan et al., 2017) with sample-weighted aggregation, FedProx (Li et al., 2020) with $\mu = 0.01$.

Personalisation (Improvement 1): FT-k post-FedAvg local fine-tuning ($k \in \{1, 3, 5, 10, 20\}$ steps, per-client learning rate selected on validation at $k = 3$), Split-FL with shared LSTM backbone and private linear output head, and Ditto (Li et al., 2021) with μ selected per client on validation from $\{0.01, 0.1, 0.3, 1.0\}$.

Personalisation (Improvement 2): SCAFFOLD (Karimireddy et al., 2020) with per-client control variates, FT-k and Ditto applied to the SCAFFOLD global model rather than the FedAvg global model, Per-FedAvg/FOMAML (Fallah et al., 2020) with inner step size 5×10^{-4} and $k \in \{0, 1, 3, 5\}$ adaptation steps evaluated at test time, FedEM (Marfoq et al., 2021) with $M = 3$ component models and temperature $\tau = 1.0$ in the E-step, APFL (Deng et al., 2020) with α grid-searched on validation from 11 values in $[0, 1]$, and pFedGraph (Ye et al., 2023) built on top of SCAFFOLD with temperature τ selected from $\{0.1, 0.5, 1.0, 5.0\}$ on seed 42.

3.6. Statistical Testing

Primary significance test: stationary block bootstrap (Politis & Romano 1994, block=21 trading days, 5,000 resamples) on daily returns pooled across 10 seeds, giving roughly 10,000 daily observations per comparison. We test one-sided (method better than baseline) for positive claims and two-sided for null results. Secondary robustness check: dual Jobson-Korkie (Memmel 2003 variance correction) and Wilcoxon signed-rank on the 10 per-seed Sharpe values, both requiring $p < 0.05$. The dual criterion is stricter because it uses only 10 observations; we report it alongside the bootstrap for completeness. The two tests agree on the sign and direction of every finding even when they disagree on statistical significance, which confirms the results are not an artefact of the testing choice.

4. Results

4.1. Baselines and Asymmetric FedAvg Benefit

Table 2 reports mean Sharpe across 10 seeds on the 2022–2026 test period. Mean-variance (B2) fails across all markets: the rolling 60-day estimation window is too noisy for 30-asset portfolios. Equal-weight (B1) is competitive on KSE-100 and Nikkei 225 because those markets had strong broad trends in the test period. Centralised training (B4) closely matches FedAvg (B5), suggesting federation successfully approximates the privacy-violating upper bound without data sharing.

Table 2. Baseline Sharpe ratios (mean across 10 seeds, test 2022–2026, 10 bps TC one-way). FedProx is identical to FedAvg at all μ values.

Method	SP500	FTSE	KSE	Nikkei
EqWt (B1)	0.633	0.567	0.849	0.888
MeanVar (B2)	0.324	0.208	−0.087	0.491
Local (B3)	0.893	0.464	0.824	0.762
Central (B4)	0.929	0.498	0.935	0.852
FedAvg (B5)	0.950	0.510	0.938	0.866
FedProx (B6)	0.950	0.510	0.938	0.867

The FedProx null result deserves explicit attention. Across all three tested μ values and all four markets, the per-market Sharpe values differ by less than 0.001 from FedAvg. This is not a tuning failure: it reflects the fact that the Sharpe loss (typically in the range -2 to $+2$) completely dominates a proximal penalty at any reasonable μ . FedProx and FedAvg are functionally equivalent for this task.

FedAvg improves over local-only (B3) at every market, but the magnitude differs substantially. KSE-100 gains +0.114 Sharpe points, Nikkei 225 gains +0.105, S&P 500 gains +0.057, and FTSE 100 gains +0.046. This ordering matches the idiosyncrasy score (Table 1), not the HD ordering, with Pearson $r = +0.71$ between idiosyncrasy and FedAvg gain. HD ordering fails because it is dominated by volatility differences (Nikkei 225 has the highest HD score but a middling FedAvg gain), whereas idiosyncrasy captures structural independence from the federation average, which is what actually predicts how much a client benefits from other clients’ gradients. Local-only seed variance is also asymmetric: Nikkei 225 std=0.184, KSE-100 std=0.131, FTSE 100 std=0.115 versus S&P 500 std=0.064. The more idiosyncratic markets are harder to train reliably without federation and gain the most when they join it.

4.2. Personalisation Methods

Table 3 and Figure 2 summarise all personalisation methods. The key pattern is consistent: methods that operate on the aggregation or training objective side (SCAFFOLD, Per-FedAvg global) raise the performance ceiling, while

methods that move parameters away from the global model in the direction of local data do not.

Table 3. All methods: mean Sharpe (10 seeds, test 2022–26). SCAFFOLD and pFedGraph compared vs FedAvg. All others compared vs SCAFFOLD. Sig: ** = both JK+Wilcoxon $p < 0.05$; . = not significant.

Method	SP500	FTSE	KSE	Nikkei
FedAvg (base)	0.917	0.517	0.949	0.917
SCAFFOLD	0.999	0.596	0.985	0.911
FT+SCAFFOLD	0.999	0.596	0.986	0.911
Ditto+SCAFFOLD	0.942	0.595	0.868	0.762
Per-FedAvg $k=0$	1.015	0.609	0.992	0.920
FedEM	0.670	0.540	0.994	0.760
APFL	0.935	0.139	0.920	0.860
pFedGraph+SC	0.996	0.585	0.984	0.909
FT-k best (D4)	0.904	0.464	0.943	0.926
Split-FL	0.784	0.069	0.931	0.666
Ditto (D4)	0.677	0.476	0.921	0.821

SCAFFOLD is the most effective intervention. S&P 500 seed std drops from 0.094 to 0.012 ($8\times$ reduction), confirming that client drift is a real source of instability. Mean Sharpe improves on S&P 500 (+0.082), FTSE 100 (+0.079), and KSE-100 (+0.036); Nikkei 225 is essentially flat (−0.006). Block bootstrap versus FedAvg: S&P 500 $p = 0.003$, FTSE 100 $p = 0.010$, KSE-100 $p = 0.072$, Nikkei 225 $p = 0.635$. The dual JK+Wilcoxon criterion does not reach significance on any market at $n = 10$ seeds, illustrating the difference in statistical power between 10 Sharpe ratios and 10,000 daily return observations.

Per-FedAvg produces the highest mean Sharpe of any method on S&P 500 (1.015) and FTSE 100 (0.609). The meta-learned global model at $k = 0$ beats FedAvg on all four markets numerically. Adding adaptation steps ($k = 1, 3, 5$) degrades performance monotonically from the $k = 0$ point on every market. The meta-objective trains a more balanced global model; adaptation away from it does not help, confirming the local optimum is no better than the global one.

FT-k and FT+SCAFFOLD. Post-FedAvg fine-tuning produces small positive deltas on S&P 500 (+0.021 at $k = 5$) and KSE-100 (+0.012 at $k = 5$), but none clear either significance criterion. FTSE 100 degrades significantly at $k = 20$ ($p < 0.05$ dual criterion). Fine-tuning from the SCAFFOLD global model adds nothing over the $k = 0$ baseline.

Split-FL significantly harms FTSE 100 (−0.394 Sharpe, $p < 0.05$ both tests) and Nikkei 225 (−0.260, $p < 0.05$ both tests). The private output head learns a local allocation that is worse than the global head for these two markets. This is a clean negative result: separating the head from the aggregation does not give these clients a useful private representation of their market-specific optimum.

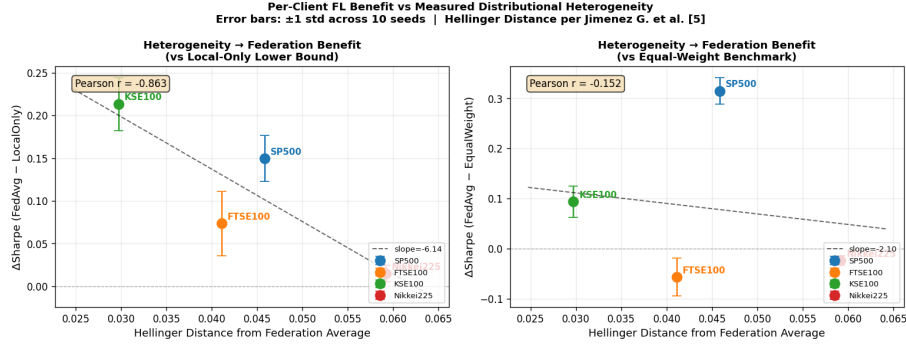


Figure 1. Per-client FedAvg gain (Sharpe vs local-only) plotted against idiosyncrasy score ($1 - R^2$). The gain is monotonically ordered by idiosyncrasy, not Hellingger Distance. Pearson $r = +0.71$.

Ditto selects $\mu = 1.0$ (fully global) for all four markets on the validation set when applied on top of FedAvg, meaning no local movement is useful. Applied to the SCAFFOLD global model, it selects $\mu = 0.01$ for three markets but degrades all four relative to SCAFFOLD. In both cases, the validation set correctly identifies that the global model is already optimal.

FedEM mixing weights collapse to near-uniform across seeds ($\lambda \approx [0.31, 0.35, 0.34]$ for S&P 500). KL divergence from uniform mixing is 0.002 for S&P 500 and 0.048 for KSE-100, indicating that the three components fail to specialise into different market regimes. Without meaningful component differentiation, FedEM reduces to a noisy version of FedAvg: S&P 500 drops 0.248 Sharpe below FedAvg.

APFL. The validation-selected local-model weight α is 0.820 for KSE-100 and 0.630 for FTSE 100. More idiosyncratic markets end up with higher local weights, which is the wrong direction economically: these are the markets that benefit most from the global signal, and APFL pulls them away from it. The result for FTSE 100 is particularly severe: $\alpha = 0.630$ causes it to collapse to Sharpe 0.139. Pearson $r(\text{idiosyncrasy}, \alpha) = +0.468$.

pFedGraph on SCAFFOLD. The gradient similarity matrix at the best temperature ($\tau = 5.0$) is near-uniform across all seeds: self-weight 0.299, peer weights approximately 0.233 each. The four markets' gradient directions in parameter space are near-orthogonal, so similarity-based reweighting carries no exploitable structure. pFedGraph matches SCAFFOLD within noise on all four markets but does not improve on it.

4.3. Regime Analysis

The test period contains three distinct phases. In 2022 (inflation shock), all methods and all markets perform poorly: KSE-100 Sharpe ranges from -1.62 to -1.79 depending on

method, reflecting a capital controls crisis in Pakistan with no historical precedent in the 2010–2019 training data. No learning-based method can protect against a policy shock of this nature. In 2023 (recovery), the federation advantage is visible: SCAFFOLD leads S&P 500 at $+2.24$ versus FedAvg at $+2.08$, and KSE-100 at $+1.87$ versus FedAvg at $+1.94$. In the 2024–2026 late-cycle phase, SCAFFOLD is the clearest leader: S&P 500 1.054 versus FedAvg 0.949, FTSE 100 1.057 versus FedAvg 0.883. Personalisation methods do not lead in any (market, regime) cell.

4.4. Phase Transition Observation

Across all eight methods, the validation-optimal personalisation parameter falls at one of two corners: fully-global (Ditto $\mu = 1.0$, Per-FedAvg $k = 0$ dominates, APFL $\alpha \approx 0$ for S&P 500) or local with worse test Sharpe ($\mu = 0.01$, high α but degraded). No market finds a useful intermediate value in any method. Figure 3 shows this across three paradigms: the region between the two corners is empty.

Liu et al. (Liu et al., 2022) prove that in the cross-silo, small- K , same-task setting, the optimal MR-MTL regulariser either goes to infinity (fully-global) or zero (fully-local), with no stable optimum between them. The intuition is that when all clients solve the same task on different data, the gradient directions are close enough in expectation that the global model already captures what each client needs, but the noise in each client's gradient is large enough that local adaptation moves away from, not toward, the true local optimum. Our results across three structurally different personalisation paradigms (regularisation-based Ditto, meta-learning Per-FedAvg, and output-mixing APFL) are mutually consistent with this prediction.

4.5. Transformer Architecture Experiment

To test whether the LSTM's recurrent inductive bias explains the results rather than the task structure itself, we trained a capacity-matched Compact Transformer: `d_model=96`,

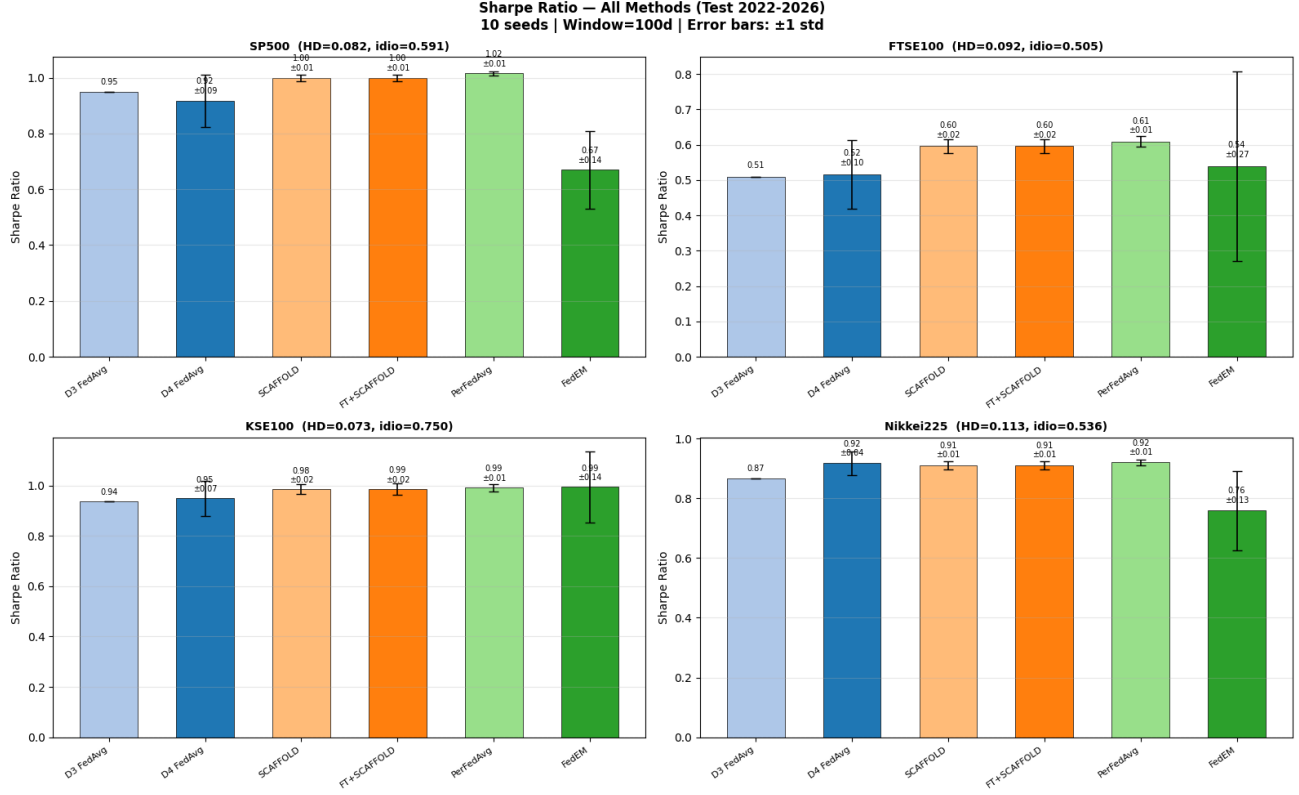


Figure 2. Mean Sharpe for all methods per market (10 seeds, error bars ± 1 std). SCAFFOLD and Per-FedAvg $k=0$ consistently lead FedAvg. Methods that adapt parameters locally do not improve on the federation baseline.

4 attention heads, 2 encoder layers, FF dim=256, Pre-LayerNorm, sinusoidal positional encoding, last-timestep pooling, 194K parameters versus the LSTM’s 184K.

Table 4. Transformer versus LSTM: local-only Sharpe (mean, 10 seeds). TF trained with Adam lr= 3×10^{-4} , patience=60, no scheduler.

Model	SP500	FTSE	KSE	Nikkei
LSTM (local)	0.893	0.464	0.824	0.762
TF (local)	0.698	−0.596	0.509	0.689

The Transformer underperforms the LSTM at every market. FTSE 100 at -0.596 indicates the model fails to learn a useful portfolio on this market even after extended training. The OneCycleLR scheduler we initially used caused the model to plateau during the warmup phase before reaching full learning rate; reducing lr to 3×10^{-4} and switching to constant Adam with patience=60 resolves this for three markets but not FTSE 100. The most likely cause is that the Transformer’s parallel attention mechanism produces high-variance gradients through the Sharpe loss on small datasets (approximately 2,300 training windows per client), whereas the LSTM’s gating mechanism naturally damps gradient

spikes across timesteps. A full Transformer-based federated study with a smoother loss function (e.g. differential Sharpe ratio) is left as future work.

5. Discussion

Why FedAvg helps asymmetrically. KSE-100 and Nikkei 225 gain more from federation because they are more structurally independent of the federation average. Their local models train on a narrower gradient signal with high seed variance (Nikkei 225 std=0.184, KSE-100 std=0.131). Federation provides a regularising effect proportional to how far each market sits from the mean gradient direction: the markets that need it most benefit the most. The S&P 500 and FTSE 100 already sit closer to the federation average, so the benefit is smaller but still positive.

Why personalisation fails. Three mechanisms combine. First, the task is structurally identical across all four clients: softmax-over-30-assets under the same Sharpe loss, architecture, and feature set. Classical personalisation methods were designed for label-skew settings where local optima are provably different. Our clients share the same optimum structure; only the data it is fit on differs. Second, the

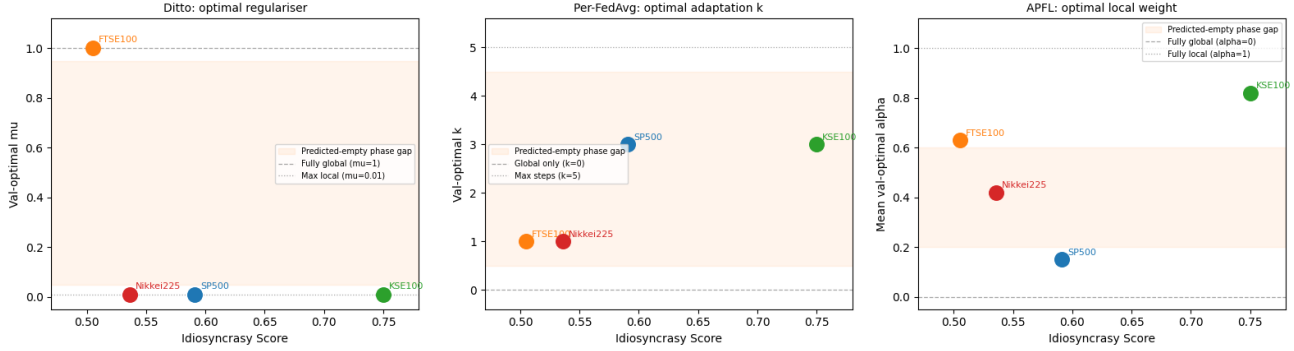
Cross-Silo Personalisation Phase Transition
(cf. Liu, Vepakomma & Smith, NeurIPS 2022 -- optimal lambda in $\{0, \infty\}$)


Figure 3. Phase transition diagnostic across three personalisation paradigms. Each panel plots the validation-optimal parameter (Ditto μ , Per-FedAvg k , APFL α) against idiosyncrasy score for each market. The orange band is the “helpful personalisation” zone that theory (Liu et al., 2022) predicts is empty in the cross-silo small- K same-task regime. It is empty across all three methods.

Sharpe loss couples all 30 weights cross-sectionally, producing a loss surface with low effective dimensionality. The directions that matter (overall volatility scaling, tail calibration, momentum versus mean-reversion balance) are shared across markets even when daily returns are uncorrelated. There is no parameter direction a single client can move in without leaving the shared basin. Third, four clients is a very small count for methods designed for tens or hundreds of clients. The pFedGraph near-uniform W matrix confirms this: gradient cosine similarity among four near-orthogonal gradient vectors carries no exploitable structure.

What does work. SCAFFOLD corrects client drift without changing the convergence point, reducing variance without harming any market. Per-FedAvg’s meta-objective shapes the global model to be a better initialisation across all clients simultaneously. Both operate on the aggregation and objective side of training rather than asking clients to move away from the global model afterward. For practitioners considering cross-market federated portfolio FL: invest in better aggregation and better global training objectives rather than per-client adaptation.

Limitations. The 30-stock universe is fixed at recent blue-chip constituents, introducing survivorship bias. FTSE 100 is sensitive to constituent selection (Sharpe changes by 23.6% under 20% random stock exclusion). KSE-100’s poor 2022 performance reflects a capital controls event with no training-period analogue, not any FL design choice. The test period covers only one full market cycle.

6. Conclusion

FedAvg produces asymmetric per-client Sharpe gains across structurally different international equity markets. The gains are predicted by an idiosyncrasy score rather than

Hellinger Distance. SCAFFOLD reduces seed variance $8\times$ on S&P 500 and is bootstrap-significant on two markets. Per-FedAvg’s meta-objective produces a better global model than plain FedAvg at zero adaptation steps. Eight personalisation methods spanning every major paradigm fail to improve any client significantly: the optimal parameter collapses to fully-global or fully-local for every client and every method, consistent with the theoretical cross-silo phase transition of Liu et al. (Liu et al., 2022). The practical implication is clear: in cross-market federated portfolio FL, improving the global training process is more reliable than trying to adapt the model per client.

Future work should test whether more markets, a Transformer with a smoother loss function, or regime-aware federation design can move this system out of the phase-transition corner and into a setting where per-client adaptation genuinely helps.

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